Teaching notes and solutions

> **Teacher note · Teacher notes.** The common test in S28 comes before the combinations of S30: include only principles and ordered lists. Test 3 may involve combinations, but not Pascal's rule taught in S35. Have students state the object being counted before any formula. For weaker students, use cards bearing names; does permuting the cards change the object? Newton's binomial theorem, cited in the scheme of work, is an extension: keep it explicitly optional.

Solutions to the activity

1. A pair is a subset of two students from six: $\binom{6}{2} = \frac{6\times5}{2} = 15$.
2. The chair can be chosen in six ways, then the secretary from the other five students: $6 \times 5 = 30$ assignments.
3. Each speech can be given to any of the six students, including one already chosen. The three successive choices give $6^3 = 216$ sequences.
4. With four cards, the pairs are AB, AC, AD, BC, BD, CD: six choices. The twelve pairs of roles are:
$$\begin{array}{lll} (A,B) & (A,C) & (A,D) \\ (B,A) & (B,C) & (B,D) \\ (C,A) & (C,B) & (C,D) \\ (D,A) & (D,B) & (D,C) \end{array}$$
There are $4^3 = 64$ sequences of three speeches, since each position has four choices.
5. For a pair, order does not matter and a student is not repeated. For the roles, order matters, with no repetition. For the speeches, order matters and repetition is allowed.

E1. There are 10^4 codes in total. Those containing no 0 have nine choices at each of the four positions, that is 9^4 codes. By complement, the number required is $10^4 - 9^4 = 3439$.

E2. There are ten choices for gold, then nine for silver and eight for bronze. The multiplication principle gives $10 \times 9 \times 8 = 720$ podiums. Changing the order changes the podium.

E3. The imposed person is already chosen. It remains to choose a subset of two people from the other six: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

E4. A path is determined by the positions of its four steps to the right among seven positions. The other three steps are then forced. We therefore count $\binom{7}{4} = 35$ paths.

E5. We must choose two girls from eight, then two boys from seven. Each pair of choices gives a single group: $\binom{8}{2}\binom{7}{2} = 28 \times 21 = 588$.

Solutions to Homework 9

1. Order does not matter: there are $\binom{12}{4} = \frac{12\times11\times10\times9}{24} = 495$ committees.
2. We choose two students from seven and two teachers from five: $\binom{7}{2}\binom{5}{2} = 21 \times 10 = 210$ committees.
3. From the 495 committees, remove the $\binom{5}{4} = 5$ committees made up only of teachers and the $\binom{7}{4} = 35$ committees made up only of students. These categories are disjoint: $495 - 5 - 35 = 455$.
4. Each of the 210 committees has two choices for the teacher chair and two choices for the student secretary: $210 \times 2 \times 2 = 840$ arrangements.
5. Sort the committees according to the number k of teachers, with $0 \leq k \leq 4$. Category k contains $\binom{5}{k}\binom{7}{4-k}$ committees. The five categories are disjoint and cover all the committees. By addition,
$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4} = 495.$$

Solutions to the textbook or past-paper work

A hand contains five distinct cards; order does not matter. There are therefore $\binom{52}{5} = 2{,}598{,}960$ hands. For a hand containing the ace of spades, this card is already chosen; we choose the other four from the 51 remaining cards. There are $\binom{51}{4} = 249{,}900$ hands. This result is not multiplied by 5: the hand has no first or last position.